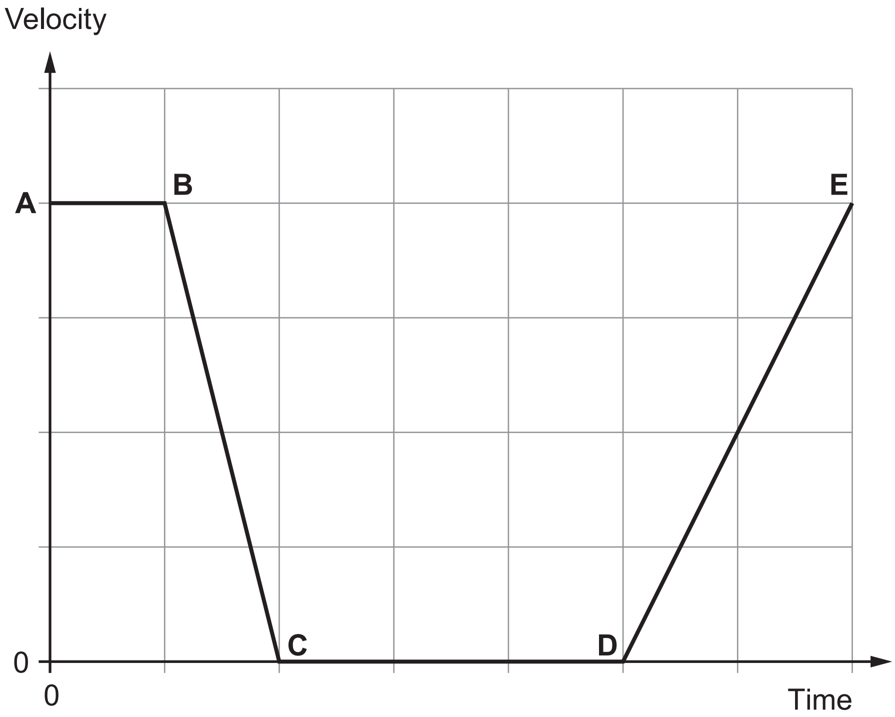
Examiner only

4. (a) Since 2017, new bi-modal electric / diesel trains have been introduced to run between Swansea and London.
A velocity-time graph for one of these trains is shown below.



Use information from page 2 to **compare** the distances travelled **and** accelerations for the parts of the journey labelled **AB**, **BC**, **CD** and **DE**. [6 QER]

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(b) The diagram shows a 5-carriage bi-modal electric/diesel train.

The table below shows information about two types of bi-modal electric/diesel trains.

Train	Mass ($\times 10^5$ kg)	Maximum speed (m/s)	Standard acceleration (m/s^2)	Standard deceleration (m/s^2)	Emergency deceleration (m/s^2)
5 carriage	2.3	55.8	0.7	1.0	1.2
9 carriage	4.4	55.8	0.7	1.0	1.2

For the journey from Swansea to London **two** of the 5-carriage trains are joined, making a 10-carriage train.
The 10-carriage train has the same speed and acceleration as a 5-carriage train.

(i) State Newton's second law of motion as an equation. [1]

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(ii) Use information from the table to answer the following questions.

I. Use an equation from page 2 to calculate the resultant force needed to accelerate the 10-carriage train. [2]

Resultant force = ... N

II. Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the time taken to accelerate the 10-carriage train from rest to its maximum speed. [2]

Time = ... s

Examiner only

(iii) A student states that the 9-carriage train is approximately **double** the mass of the 5-carriage train and if they are both travelling at the same speed it will take approximately **double** the time to stop in an emergency.
Explain whether the **2 claims** made by the student are correct. [3]

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